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Chapter 3

Material Requirements Planning



Material Requirements Planning

Introduction

Material Requirements Planning is a system for planning the production and procurement of components and materials in order to satisfy future demand. It plans supplies for components and materials in terms of items, quantities and release dates. It sets priorities for both production and purchasing by determining due dates for supplies. It provides the input for full capacity planning.

MRP converts suggested MPS orders, forecasts and any other demand into a detailed schedule. The inclusion of MPS suggested orders is optional. The detailed schedule satisfies the calculated demand according to the planning rules, which you establish.

The schedule includes recommendations for generating new suggested orders/schedules and for changes to be made to existing orders/schedules.

To include items in the MRP planning run, they are not normally flagged as MPS items in the Production Items Detail file. However, the system includes MPS items if they are dependent on MRP items but this situation should not normally be allowed to occur.

There is no equivalent of rough cut capacity planning in the MRP application because it is assumed that having reached this detailed level of planning, full capacity planning is undertaken.

Material Requirements Planning Introduction - Cellular Planning

MRP is able to identify the material requirements and planning supplies in a Production cell. When an MRP plan is produced it is possible to view that plan from the perspective of a Cell Manager in order to show the material supply and demand for which the manager has responsibility. A cell is defined by the Inventory Department code as set up in the Inventory Descriptions file. A stockroom is assigned to a department.

Department codes can also be created on the Production parameters file within the user parameter type DEPT for definition against work stations during maintenance (see Work Station) and on the Inventory Description file for type code DEPT.



Note: In Stockroom Site, only 3 characters are allowed for this field.

After a cellular MRP planning run, the MRP Enquiry and Reports can be selected by Department (Cell). Supply and Demand is balanced at the Item/Stockroom level rather than just at item level.

The system generates a supply record and a departmental demand record for both a demand stockroom and a supply stockroom. Pegging facilities provide extra visibility for a Cell Manager to see which other cells are providing the demand on his/her cell and which cells are being supplied by his/her cell.

Considerations:

- This is an MRP function only.
- Each MRP run must be either Cell based or Model based.
- MRP forecasts are not extracted at this time.

- There are not any work station schedules by cell.

Material Requirements Planning Introduction - MRP Processing

To run MRP, you must first maintain Model Stockroom and MRP Reporting Profile Maintenance in the same way as MPS. The stages of MRP processing may be summarised as follows:

- Maintain MRP Reporting Profile (once only)
- Maintain Model Stockroom (once only)
- Maintain Net Change Reason codes
- Maintain Seasonal Indices *
- Maintain Product Families *
- Maintain Family Forecasts *
- General Item Forecasts *
- Maintain Item Forecasts *
- Spread Item Forecasts *
- RUN MRP
- Make Enquiries and run reports
- Confirm MRP Suggested Orders/Schedules
- Create Work Station Schedules

* If MRP forecasting is to be used.

Material Requirements Planning Introduction - MPS/MRP Common Features

Certain tasks are common to both the Master Production Scheduling and the Material Requirements Planning systems. These are listed below:

- Maintain Model Stockrooms enables new models to be defined with appropriate stockrooms. For more information see Maintain Model Stockrooms.
- Maintain MRP Reporting Profile sets out the time periods to be used to supply the schedules. For more information see Maintain MPS Reporting Profile.
- Forecasting allows forecasts to be set up to create demand for planning purposes.



Note: In order to save replication of data, the above common features are only documented once.

Maintain Model Stockrooms



1/MRP

This is the same as Maintain Model Stockrooms for MPS.



Select Maintain Model Stockrooms on the menu.

For more information see Maintain Model Stockrooms in Master Production Scheduling.

Generate Model Primary Stockrooms

1...
2...

2/MRP

This is the same as Generate MPS Model Primary Stockrooms.

Maintain MRP Reporting Profile



3/MRP

This is the same as Maintain MPS Reporting Profile.



Select Maintain MRP Reporting Profile on the menu.

For more information see Maintain MPS Reporting Profile in Master Production Scheduling.

Maintain Net Change Reason Codes

1...
2...

4/MRP

MRP may be run in three different ways, these are fully explained in the next section.

One of these ways is called a Net Change MRP run. This kind of MRP run is restricted in its update to just certain specified events which have taken place since the last Net Change MRP run.

These specified events are viewed as Reason codes. The Reason codes to be considered for any Net change MRP run can be maintained in advance by the user. This option allows amendments to be made if required.

When an event takes place within the manufacturing system for which a Reason code has been selected, a trigger is created. For example, if the Reason code for Item Master Maintenance has been selected and an update is made to one item, then a trigger is set for this one item by the system. Each time there is an update to the Item Master Maintenance then a new trigger is created. Therefore it is possible for many triggers to be set for any Reason code.

When a Net Change MRP run is requested the number of triggers set against each selected Reason code can be viewed if required.



Select Maintain Net Change Reason Codes on the menu.

Code	Description	Active (1=Yes,0=No)
01	Forecast v Actual Comparison	1
02	Forecast Generation	1
03	Item Forecast Maintenance	1
04	Item Master Maintenance	1
05	Unplanned Inventory Trans.	1
06	Requisition Maintenance	1
07	Unplanned Issue or Receipt	1
08	Purchase Order Maintenance	1
09	Order Maintenance	1
10	Sales Order Maintenance	1

F3=Exit F8=Update F12=Previous

08-15 SF MK KS IM II KENB20A PET166S2

Fields:

Active

1 - To select a reason code.

0 - To switch off a reason code



Note: This option must be taken before attempting any Net Change MRP run.



Note: The descriptions of the Reason codes can be maintained in the Parameter File using code

TRIG, if the user is authorised.

Maintain Net Change Reason Codes - Reason Code Definitions

Reason code	Description	Option
01	Forecast Vs Actual Comparison	Future Development
02	Forecast Generation. Trigger set by generation of MRP forecasts from the Forecasting application.	Forecasting (not in MPS or MRP)
03	Item Forecast Maintenance. Change family forecast in MRP and generate or maintain item forecast in MRP	Maintain Family Forecast Generate Item Forecast Maintain Item Forecast Spread Item Forecast
04	Item Master Maintenance	Production Details
05	Unplanned Inventory Transaction	Miscellaneous Receipt Miscellaneous Issue (Physical) Miscellaneous Issue (Available)
06	Requisition Maintenance. Requisitions are not extracted in planning schedules	Requisitions Confirmation
07	Unplanned Issue or Receipt. Over issue or receipt against a production order.	Issue Materials
08	Purchase Order Maintenance	Order Entry Order Amendment
09	Order Maintenance. Change of date or quantity on a production order.	Maintain Orders
10	Sales Order Maintenance. Create or maintain sales orders	Order Entry (Conversational) + others
11	Route/Structure Maintenance	Routes/Structures
12	MPS Requirements Maintenance. Change to a production order for MPS parent with MRP components.	Maintain Orders
13	Available to Ship Exception	Available To Ship Report
14	MPS Schedule Amendment	MPS Review Line Schedule Maintenance
15	MRP Schedule Amendment	MPS Review Line Schedule Maintenance



Note: To generate a trigger the changes must be in the stockroom related to the model being used for the net change run.

Maintain Supply Sourcing Rules



5/MRP

This option is the same as Maintain MPS Supply Sourcing Rules

Run MRP



11/MRP

The Material Requirements Planning run is a series of calculations undertaken by the system, using data derived from the other applications in the Production system, to construct a schedule of possible future production requirements.

This can only be done when model maintenance has been completed. If required, forecast details can also be entered, before the run is started. MRP may be run in a number of different ways to reflect your planning requirements.

Full Regenerative - All items defined against stockrooms within a model are included

Net Change - Only items which meet certain criteria (triggers) are included. These triggers are specified using a separate option in MRP maintenance.

Selective - Only items within specified ranges are included.

The MRP processing factors are the same as the MPS Processing Factors already described. However, there are of course differences in the items selected. The following items are considered during the MRP run:

- MRP items, which are held in a stockroom defined in the model.
- MRP items which have been triggered (for a Net Change run).
- MRP items which meet user selection criteria (for a Selective run).
- MPS items which are parents of MRP items held in stockrooms defined in the model.

All the other factors operate in the same way as in MPS.

This application takes the selected planning definition and extracts all the stockrooms defined. The stockrooms define the scope of MRP. Only items that reside in the stockrooms on the model are extracted for planning.



Note: It is important to ensure that calendars have been maintained to cover the duration of the review.



Select Run MRP on the menu to display the first MRP Selection screen.

MRP Selection Screen 1



Select Run MRP on the menu to display the first MRP Selection screen.

Fields:

Model

Enter the MRP model to run. The stockrooms defined to the model determine the scope of inclusion of items within the run. Only one model, defined on the company profile, may be used to confirm any suggested production orders or schedules. If such suggestions are to be confirmed, it is essential that the live model code is entered here.



Press **Enter** on the first MRP Selection screen to display the second screen.

MRP Selection Screen 2



Press **Enter** on the first MRP Selection screen to display the second screen.

MR002	JW - Multiplant Phase II QA Company	USER GBUJAJFW	5/02/98 11:36:09
MRP Run Options		MRP Model : CM Cental Model	
		Production Calendar : 01	
Details shown are from the last run selection.			
MRP Run Type.....		1 = Regenerative	
		2 = Net Change	
		3 = Selective	
Enter: Current Date.....		260198	
Overdue Days.....		---	
Time Fence Days.....		---	
Safety Horizon Days.....		---	
End Date.....		311298	
Suppress low level code generation.. 1 (1=Yes)			
F3=Exit	F12=Previous	F16=Review Selection	F17=Start Run
F18=Additional Parameters		F22=Print Details	
8-42	SA	MW KS CL IM DM II	EDN400A KB PET165S2

Fields:

MRP Run Type

- 1 - Full Regenerative. All items defined by the model are reviewed.
- 2 - Net Change. Items in the model definition which have been triggered as changed are processed. (**F16=Review Selection**)
- 3 - Selective. Items in the model definition which meet the run time selection criteria are processed. (**F16=Review Selection**)

Current Date

This is the base date for the review. The review start date and the frozen schedule time fence are calculated from this date.

Overdue Days

Enter the number of days of overdue supply and demand that you wish the MRP to consider. This number of days is subtracted from the current date to give the MRP start date. All supplies and demand due on or after this date is extracted.

Time Fence Days

Enter the number of days that form the frozen portion of the schedule. The application does not make any recommendations to schedules due within the time fence or suggest new orders within it. The time fence is calculated as the current date plus the specified number of days. The time fence period is typically the longest manufacturing or procurement lead time of your items.

Safety Horizon Days

This a planning factor to extend the horizon of each items plan in order to ensure that lower level generated demand falls within the overall plan.

End Date

Enter the end date for the MRP review. This date should represent the longest cumulative lead time of your MRP items, plus a safety factor. This date is over-ridden by individual item end dates if that option is selected.

Suppress Low Level Code Generation

Suppress low level code generation indicates whether or not low level codes should be generated as part of the MRP run. This routine can be omitted if no changes have been made to any of the product structures; or the standalone Generate Low Level Codes option in Utilities is run prior to running MRP.



If 2 (Net Change) is entered in the MRP Run Type field, select **F16=Review Selection** to display the Net Change Trigger Selection window.



If 3 (Selective) is entered in the MRP Run Type field, select **F16=Review Selection** to display the Enter Selection Criteria window.

Functions:

F16

Review Selection. If 1 (Regenerative) is entered in the MRP Run Type field, this function is not available. If 2 (Net Change) is in the MRP Run Type field, press **F16** to display the Net Change Trigger Selection window. If 3 (Selective) is in the MRP Run Type field, press **F16** to see the Enter Selection Criteria window.

F17

Start Run. The selected information is submitted and the MRP run begins.

F18

Additional Parameters. For more information see Additional Parameters Screen (F18).

F22

Print Details. Printout a summary listing of the planning parameters.

Type field on the MRP Selection screen, select

F16=Review Selection to display the Net Change Trigger Selection window.

```

MRP Run Type..... 2      1 = Regenerative  2 = Net Change
Net Change Trigger Selection
Sel Trigger Description                               Unprocessed Quantity to
Reason Triggers Process
1 01 Forecast v Actual Comparison                     0 999999999
1 02 Forecast Generation                             0 999999999
1 03 Item Forecast Maintenance                       0 999999999
1 04 Item Master Maintenance                         6 999999999
1 05 Unplanned Inventory Trans.                     0 999999999
1 06 Requisition Maintenance                       0 999999999
1 07 Unplanned Issue or Receipt                     0 999999999
1 08 Purchase Order Maintenance                     0 999999999
1 09 Order Maintenance                             0 999999999
1 10 Sales Order Maintenance                        0 999999999
-----
1=Select All  2=Select with Limit
F3=Exit  F12=Previous  F18=Select All

```

Trigger Reason

Unprocessed Triggers

Sel

- ### Quantity To Process

Functions:

F18

Select All. Select all reasons to process.

Enter Selection Criteria Window (F16)



If 3 (Selective) is entered in the MRP Run Type field on the MRP Selection screen, select **F16=Review Selection** to display the Enter Selection Criteria window.

Enter the ranges to be used to select items for processing. An item is only selected for review if it matches all the selections entered. If no selections are made, all items are processed; if this is intentional, then a regenerative run is more efficient.

```

MR002          Z1 - GB Demonstration Company      USER GBJBAMS      8/12/95
                                           13:49:47

MRP Selection      MRP Model      : 01 model 01
                   Production Calendar : 01

MRP Run Type..... 3    1 = Regenerative  2 = Net Change
                       3 = Selective

Enter Selection Criteria

Item.....?..      to
Planner.....?..    MS      to      MS
Item Group.....?..  to
Value/Usage Class.... to
Preferred Supplier.... to
GT Family..... n    to

F4=Prompt  F12=Previous
F22=Print details
11-30      SF      MK      KS      IM      II  KENB20A  PET166S2
  
```

Fields:

Item

Enter the range of items to be reviewed.

Blanks indicate no selection by Item Code.

Planner

Enter the range of planners to be reviewed. Only items for this range of planners are processed.

Blank indicates no selection by planner.

Item Group

Enter the range of Item Groups (Product Families) to be reviewed. Only items for this range are processed and made available on enquiries and reports. Blank indicates no selection by Item Group. Item Groups are categorised by the Product Group Minor Code in Inventory Management.

Value/Usage Class

Enter the range of classes to be reviewed. Only items with the selected range of Value/Usage Class are processed.

Blank indicates no selection by Value/Usage Class. This value is maintained by Inventory Management.

Preferred Supplier

Enter the range of Preferred Suppliers to be reviewed. Only items associated with the requested suppliers are processed. Blank indicates no selection by Preferred Supplier.

GT Family

Enter the range of GT Family codes required.

Additional Parameters Screen 1 (F18)



Select **F18=Additional Parameters** to display the first of two additional parameter screens..

```

MR017          JW - Multiplant Phase II QA Company      USER GBUJBAJFW      5/02/98
                                                    11:36:27
Additional Parameters  MRP Model          : CM Central Model
                      Production Calendar: 01              (1 of 2 )

Planning Basis..... ☒ 0 = Standard
                      1 = Multi-plant Network
                      2 = Single Unit Sourcing
                      3 = Cellular Network

Planning Unit Parameters
Use Item Time Fence..... _ (1=Yes)
Use Item End Date..... _ (1=Yes)
Dependent Requirements Basis..... _ (0=Suggested, 1=Firm)
Forecast Basis..... _ (0=Daily, 1=Period)
Confirmed Schedule Policy.....?. _

(If multiple planning units are planned by a central run model, the
parameters stated above are extracted from the individual unit definitions.)

F3=Exit  F4=Prompt  F8=Update  F12=Previous

-----
6-37      SR      MW      KS      CL  IM  DM  II      EDN400A  KB  PET165S2
  
```

Fields:

Planning Basis

You can choose between the following:

- 0 - Standard Run. This selection operates without Multiplant facility, that is, the MRP run is processed at organisational level where all load, resources and capacity are considered.
- 1 - Multi-Plant Network. This incorporates the concept of multiple planning units that can be considered either as a related set of manufacturing centres supporting an overall manufacturing plan, or as autonomous units responsible for their own supply and demand situations without reference to other planning units or a higher level plan.
- 2 - Single Unit Sourcing. Allows a plan to be developed which satisfies demand by allocating recommended supplies for an item to different manufacturing routes within a single planning unit, without reference to any other planning unit.
- 3 - Cellular Network. Cellular Network Planning provides an optional third tier of planning that can be defined in the planning structure. This type of planning can be processed independently, or as a related group within a Central or Plant Model.

Use Item time Fence

As an alternative to a global time fence, a time fence based on each items production lead time may be used.

0 - Use time fence entered for the run. If no lead time exists for the item, the run time fence is used.

1 - Use item time fence.

Use Item End Date

In preference to a global end date for the review of each item, an end date based on each item's cumulative lead time may be used.

1 - To use item end dates.

0 - To use end date entered for run as a global value.

If no cumulative lead time exists for an item, the global end date is used.

The item end date is calculated as:

Current Date + Cumulative lead time + Safety Horizon days.

For non-production items, the purchase lead time is used for both production and cumulative lead times.

Dependent Requirements Basis

Indicates on what basis dependent demand is calculated.

0 - Use only suggested parent item quantities to determine dependent demand, including suggested changes for firm orders/schedules.

1 - Use only firm parent quantities, to determine dependent demand ignoring any suggested changes.

Forecast Basis

This option controls which date is used as the demand date on extracted forecasts.

0 or Blank - Use the forecast date.

1 - Use the end date of the forecast.

Confirmed Scheduled Policy

You can enter the following options:

0 - Exclude. Recalculate the whole MPS plan.

1 - Include. Include any confirmed manufacturing schedules in the plan.

The default is taken from the planning option section within Company Profile.



Select **F8=Update** to validate the data you have entered and return to MRP Selection Screen 2.



Press **Enter** on the first Additional Parameters screen to display the second one.

Additional Parameters Screen 2 (F18)



Press **Enter** on the first Additional Parameters screen to display the second one.

```

MR017          JW - Multiplant Phase II QA Company      USER GBUJBAJFW      5/02/98
                                                    11:36:50
Additional Parameters  MRP Model          : CM Cental Model
                      Production Calendar: 01              (2 of 2 )

Run Model Parameters
Include Suggested MPS..... 2 (1=Yes, 2=With Phantoms)
Daily Summaries For Co-products. - (1=Yes)
Forecast Netting Basis..... - (0=Central, 1=Planning Unit, 2=Both)
Multi-sourcing Basis..... - (0=Ignore, 1=Central+Plan.Unit, 2=P.Unit)
Save Critical Resource Load..... (1=Yes)
Critical Resource Load Policy... - (0=Refresh, 1=Alternative Model Profile,
                                   2=Alternative Model With Update)
Alternative Resource Model..... CM

(The parameters stated above are only considered by the quoted run model.)

F3=Exit F8=Update F12=Previous

7-36  SA  MW  KS  CL  IM  DM  II  EDN400A  KB  PET165S2

```

Fields:

Include Suggested MPS

This selection enables you to include suggested MPS schedules from the corresponding MPS plan as an input to the material planning run. This option is only available with the regenerative run.

- 1 - To include suggested MPS schedules.
- 2 - To include any MPS phantom items.

Daily Summaries for Co-products.

- 0 - No. Option is not selected.
- 1 - Yes. This option enables the system to provide the information it uses to select co-products.

Forecast Netting Basis

You use this to determine the way in which multiplant forecasts are processed in planning.

- 0 - Central. Select this if you only want the forecast for the central model to be included in calculations.
- 1 - Planning Unit. Choose this option to pick up forecasts for planning units (plants and or cells) planned within a central model.
- 2 - Both. Incorporate both the previous 0 and 1 options.

Multi-sourcing Basis

You use this to inform MRP of which multisourcing rule it needs to apply.

0 - Ignore. Enter this if you want multisourcing rules to be ignored, that is, you require single sourcing.

1 - Central + Plan. Unit. Use this to consider rules for the central model set up with subsets of planning units, for example, central and plant models.

2 - P. Unit. Planning Unit only considers those multisourcing rules which are defined for the planning unit.

Save Critical Resource Load

1 - Yes. Use this to record the load against a critical resource so that it can be analysed later to see if its capacity has been exceeded.

Critical Resource Load Policy

You use this to define how critical resources are to be used. There are three policies:

0 - Refresh. Each run of the model will extract the critical resource capacity from the capacity definitions created by Maintain Critical Resources and Maintain Capacity Profiles. The resource is loaded into the time buckets defined on the run model's reporting profile. No load is initially included and the total capacity is available from the start.

1 - Alternative Model Profile. If you select this option, you must enter a model in the subsequent Alternative Resource Model field. This policy differs from the previous one in that an alternative reporting profile can be nominated. The reporting profile is derived from the alternative resource model. If existing load is present on the alternative model then, as long as it is not as a consequence of supply relating to the run model, it is included in the extract.

2 - Alternative Model With Update. By choosing this policy you must enter a model in the next Alternative Resource Model field. The difference with this policy is that the run model and the alternative model will be updated with the current critical resource balances when the run is completed.



Note: This field is only applicable if multisourcing is defined as a requirement in the production company profile.

Alternative Resource Model

You must enter a model in this field if you choose options 1 or 2 in the previous Critical Resource Load Policy field.



Note: This field is only applicable if multisourcing is defined as a requirement in the production company profile.



Select **F8=Update** to validate the data you have entered and return to MRP Selection Screen 2.

Review MRP

The MRP Review takes the same form as the MPS Review. For more information see Review MPS.



12/MRP



After an MRP run, select Review MRP on the menu and enter the model you wish to enquire upon. Press **Enter** to input further selection. Enter the item, cell or planner options required on the enquiry selection screen and press **Enter** to display the Enquiry Summary screen.

Enquiry Summary Screen



After an MRP run, select Review MRP on the menu and enter the model you wish to enquire upon. Press **Enter** to input further selection. Enter the item, cell or planner options required on the enquiry selection screen and press **Enter** to display the Enquiry Summary screen.

MR050	JW - Multiplant Phase II QA Company	USER GBJBAJFW	11/02/98 10:34:29
Enquiry - Summary			
Start 8/02/98		Time Fence 9/02/98	
F/C Time Fence 9/02/98		End 30/06/98	
Item....	CD-ASS6A	SUB-ASSEMBLY A FOR ITEM CD-ASS6	
Cell....	J6 Autonomous Plant 6	MPS Item	
Planner..	CD	Planning Basis: Cellular Network	
GT Family....	Product Group..	UOM EA	
		Last Planned.... 11/02/98	
	Opening Stock..		
	Safety Stock..	.000	
	Aug.Wkly Usg..	.000	
	<u>OVERDUE</u>	<u>09/02/98</u>	<u>TOTAL</u>
Actual Dem.	15000	15000	
Forecast...			
Adj.Demand.	15000	15000	
Supply....			
Available..	15000-		
Net Reqmt..	15000		
Plan Recpt.			
Plan Avail.	15000-		
F3=Exit	F12=Previous	F16=Previous Item	F17=Detail F24=More Keys
1-1	SA	MW KS CL IM DM II	EDN400A KB PET165S2

Fields:

Item

The item selected for MRP review.

Cell

The cell code and description are displayed.

Planning Basis

Displays the planning basis from the following :

- Standard run.
- Multiplant.
- Single unit.

For more details see Run MRP.

Planner

Displays the planner associated with the item.

Product Group

If the item belongs to a product group then it is displayed here.

UOM

The item's unit of measure.

GT Family

The group technology code is displayed, if present.

Last Planned

The date the item was last included in an MRP run.

Opening Stock

Shows the total Inventory quantity of the item in all stockrooms.

Safety Stock

Shows the total Inventory re-order point quantities of this item in all stockrooms within the selected cell.

Avg.Wkly Usg.

The average weekly usage. Shows the total average usage of the item in all stockrooms within the selected cell.

Actual Demand

The total demand for the item split between the specified time periods.

Forecast

The total forecast for the item, represented within the specified time periods.

Adj.Demand

The adjusted demand if applicable.

Supply Available

The total amount of the item available is shown in each time period after the adjusted demand is considered.

Net Reqmt

The supply needed to meet the net demand for each time period.

Plan Recpt

The anticipated receipt of the item based upon existing purchase and production orders.

Plan Avail

The difference between the planned receipts and net requirements.



Select **F17=Detail** on the Enquiry Summary screen to display the Enquiry Detail screen for the item.

Functions:**F16**

Previous Item. Select this if you want to see details of the item immediately before the one currently displayed.

F17

Detail. If you select this function, the Enquiry Detail screen is displayed.

F19

Next Item. Select this if you want to see details of the item following the one currently displayed.

F20

Next Planner.

Enquiry Detail Screen (F17)



Select **F17=Detail** on the Enquiry Summary screen to display the Enquiry Detail screen for the item.

MR050	JW - Multiplant Phase II QA Company	USER GBJBAJFW	11/02/98
			10:43:10
Enquiry - Detail Planning Model Autonomous Plant 6			
Start	8/02/98	Time Fence	9/02/98 F/C Time Fence 9/02/98 End 30/06/98
Item....	CD-ASS6A	SUB-ASSEMBLY A FOR ITEM	CD-ASS6 MPS Item
Cell....	J6 Autonomous Plant 6	Planning Basis:	Cellular Network
MRP	<--- Demand ---> <----- Supply ----->		
0	Date Ref	Quantity St Supply	Quantity St Action Available
■	9/02	3000 SW	3000-
-	16/02	3000 SW	6000-
-	23/02	3000 SW	9000-
-	2/03	3000 SW	12000-
-	9/03	3000 SW	15000-
Bottom			
2=Maintain Supply	3=Display All Outputs	5=Maintain Order	6=Peg Demand
F3=Exit	F5=Refresh	F12=Previous	F23=More Options
F10=New Supply	F14=Hide Exc.Demand	F15=Exc.Zero Supply	F24=More Keys
9-2	SA	MW	KS CL IM DM II EDN400A KB PET165S2

Fields:

O

Option. Enter one of the following options:

- 2 - Maintain Supply.
- 3 - Display All Outputs.
- 5 - Maintain Order.
- 6 - Peg Demand.
- 7 - Supply Details. The Supply Details window is displayed.
- 8 - Demand Details. The Demand Details window is displayed.



Enter 7 in the O field on the Enquiry Detail screen to display the Supply Details window.



Enter 8 in the O field on the Enquiry Detail screen to display the Demand Details window.

Functions:

F23

More Options. Select this function to display options 7 (Supply Details) and 8 (Demand Details) on the Enquiry Detail screen.

Peg Demand Details Window (Option 6)

Select this option to identify where the demand for the item originates from.



Enter 6 in the O field on the Enquiry Detail screen to display the Peg Demand Details window.

MR050	JW - Multiplant Phase II QA Company	USER GBJBAJFW	11/02/98
Enquiry - Detail	Planning Model Autonomous Plant 6		11:10:09
Start 8/02/98			
Item.... CD-ASS6	PARENT ITEM CD FOR PLANT J6	MPS Item	
	Planning Basis: Cellular Network		
MRP <---- Demand ---->	<----- Supply ----->		
0 Date Ref	Quantity St Supply	Quantity St Action	Available
- 13/02		Route B 1000 SW	1000
- 15/02	1000 FC		0
- 20/02		Route B 1000 SW	1000
- 22/02	1000 FC		0
- 27/02		Route B 1000 SW	1000
- 1/03	1000 FC		0
- 6/03		Route B 1000 SW	1000
- 8/03	1000 FC		0
- 13/03		Route B 1000 SW	1000
- 15/03	1000 FC		0
Bottom			
2=Maintain Supply	3=Display All Outputs	5=Maintain Order	6=Peg Demand
F3=Exit	F5=Refresh	F12=Previous	F23=More Options
F10=New Supply	F14=Hide Exc.Demand	F15=Exc.Zero Supply	F24=More Keys
9-2	SA	MW	KS CL IM DM II EDN400A KB PET165S2

Fields:

O

Option. Enter one of the following options:

- 2 - Maintain Supply.
- 3 - Display All Outputs.
- 5 - Maintain Order.
- 6 - Peg Demand.
- 7 - Supply Details. The Supply Details window is displayed.
- 8 - Demand Details. The Demand Details window is displayed.



Select **F12=Previous** to return to the Enquiry Details screen.

Cell Supply Details Window (Option 7)

Select this option to see which stockroom and cell supplies the item. This is used in a multiplant environment.

- ➡ Enter 7 in the O field on the Enquiry Detail screen to display the Cell Supply Details window.

```

MR050          Z1 - GB Demonstration Company      USER GBJBAMS      8/12/95
                                           14:04:36

Enquiry - Detail      Planning Model 01 model 01
Cell. XX Department X Start 26/11/95 Time Fence 4/12/95 End 30/05/96
Item..... MS004      Product C                  MRP Item

MRP <---- Demand ----> <----- Supply ----->
0 Date Order Quantity St Supply Quantity St Action Available
1/01          14 FX Route 1          6 SW                      6

Cell Supply Details

Supply Stockroom: FG Finished Goods      Supply Cell: XX Department X
Demand Stockroom: FG Finished Goods      Demand Cell: XX Department X

F12=Previous

- 1/04          17 FC Route 1          17 SW                      8
- 1/04          8 FS                      8
- 6/05          22 FC Route 1          23 SW                      9
                                           More...

7=Cell Supply Details      8=Cell Demand Details
F3=Exit      F5=Refresh      F12=Previous      F23=More Options
F17=Summary      F18=Header Details      F19=Next Item      F24=More Keys
13-02      SF      MK      KS      IM      II KENB20A PET166S2
  
```

- ➡ Select **F12=Previous** to return to the Enquiry Details screen.

Cell Demand Details Window (Option 8)

Select this option to see which stockroom and cell provides the demand for the item. This is used in a multiplant environment.



Enter 8 in the O field on the Enquiry Detail screen to display the Cell Demand Details window.

MR050		Z1 - GB Demonstration Company		USER GBJBAMS		8/12/95 14:06:02	
Enquiry - Detail				Planning Model 01 model 01			
Cell. XX Department X		Start 26/11/95		Time Fence 4/12/95		End 30/05/96	
Item..... RAM04		Raw material 4		MRP Item			

MRP <---- Demand ----->		<----- Supply ----->	
0	Date Order	Quantity St	Supply Quantity St Action Available
4/10		314 SW	

Cell Demand Details

Demand Stockroom: RM Raw Materials..Gen Parent Cell:

Supply Stockroom: Supply Cell:

F12=Previous

-	7/12	92 SW A000124	346 CP RI07/12 CQ	92	0
-	8/12	76 SS A000126	33 CP RI08/12 CQ	76	0
-	11/12	8 SW A000125	33 CP RI11/12 CQ	8	0

More...

7=Cell Supply Details		8=Cell Demand Details	
F3=Exit	F5=Refresh	F12=Previous	F23=More Options
F17=Summary	F18=Header Details	F19=Next Item	F24=More Keys
16-02	SP	ML	KS IM II KENB20A PET166S2



Select **F12=Previous** to return to the Enquiry Details screen.

Review Critical Resource Load



13/MRP

Use this option to display the capacity and critical resource load of plants and cells within a model. The availability of the critical resource is also shown and is represented in the unit of measure defined in Critical Resources Maintenance.



Select Review Critical Resource Load to display the first Review Critical Resource Load screen

Fields:

Model

Enter the run model. The **F4=Prompt** is available.



Press **Enter** to display the second screen.

Review Critical Resource Load Screen 2



Press Enter on the first Review Critical Resource Load Screen to display the second screen.

```
DB586          JW - Multiplant Phase II QA Company  User GBJBAJFW  12/02/98
                                                    11:33:53

Review Critical Resource Load

Model.....: CM Cental Model                      Type.....: Central
Planning Basis.....: Multi-plant Network             Start Date:  8/02/98
Cell.....?:  03 Plant 3                             End Date..: 30/06/98

Critical Resource.?: 1B Plant 2 Labour
Capacity UOM.....: HR Hours                        Calendar Code...: 01

                OVERDUE      09/02/98              TOTAL
Capacity.....          30600.000          30600.000
Load.....
Available.....          30600.000          30600.000
Load on Cell.....

F3=Exit  F4=Prompt  F12=Previous

7-23  SA  MW  KS  CL  IM  DM  II  EDN400A  KB  PET165S2
```

Fields:

Cell

You have the option to enter a cell ID if you do the load on that specific cell is displayed.

Critical Resource

Enter the critical resource required here.



Select **F12=Previous** to enquire on another model.

Confirm MRP Suggested Orders



14/MRP

The recommendations consist of purchase and production orders. If Purchase Management has been installed, the suggested purchased orders can be converted into full purchase orders, using the Requisitions system.

This screen displays all suggested production orders generated from the MRP model, which have a starting date falling within the selected range.

The selected orders can be processed, and confirmed orders for the suggested due dates and quantities can be generated. The designated planning route defined in the Item Master file is used to generate operation and component details. The status of the suggested order on the planning file changes to confirmed.

All orders displayed are automatically selected for processing. To prevent an order from being processed, remove the 1 against it.



Select Confirm MRP Suggested Orders on the menu.

Fields:

Model

The live planning model defined on the company profile is displayed.

Start Date Range

Enter the From Date and To Date, to restrict the confirmation of suggested production orders to those with a start date within the selected range. Leave these fields blank to include all suggested production orders within the model.

Planner Code Range

Enter the From and To planner codes to be selected for this review of suggested orders.

MR052	Z1 - GB Demonstration Company	USER GBJBAMS	8/12/95 14:09:17
Order Creation Model.. 01 model 01			
Start Date Range... 26/11/95 To 30/05/96 MRP Start. 26/11/95 MRP End. 30/05/96			
0	Start	Quantity	Item Description
1	1/01/96	6	MS004 Product C
1	24/01/96	71	MS004
1	2/02/96	23	MS004
1	5/02/96	21	MS004
1	4/03/96	18	MS004
1	1/04/96	17	MS004
1	6/05/96	23	MS004
1=Create Order Blank/0=Do Not Create Order 3=Display All Outputs			
F3=Exit F8=Update and Submit Job F12=Previous			
06-02 SF MW KS IM II KENB20A PET166S2			

Fields:

O

Option. You can enter the following options:

0 (or blank) - Do Not Create Order.

1 - Create Order.

3 - Display All Outputs.

Confirm MRP Suggested Schedules



15/MRP

This option enables you to see a list of suggestions for supply to meet the demand and, if required, to choose certain suggested schedules to be retained as confirmed schedules to be taken into account by future MRP runs.

Any suggested schedules which have not been confirmed, are deleted on the next MRP run.



Note: During the confirmation of work station schedules (see Confirm Work Station Schedule), a flag may be set to confirm related MPS/MRP schedules.



Select Confirm MRP Suggested Schedules on the menu to display the first Confirm Suggested Schedules screen.

Confirm Suggested Schedules Screen 1



Select Confirm MRP Suggested Schedules on the menu to display the first Confirm Suggested Schedules screen.

Fields:

To Date

Enter a to-date.

Planner Range

Enter a planner code range if required.

Item Range

Enter an item range if required.



Press **Enter** on the first Confirm Suggested Schedules screen to display the second screen.

Confirm Suggested Schedules Screen 2



Press **Enter** on the first Confirm Suggested Schedules screen to display the second screen.

MR059		Z1 - GB Demonstration Company		USER GBJBAMS	8/12/95
					14:09:49
Confirm Suggested Schedules					
Production Model		01 model 01			
Sel	Item	Due Date	Schedule Qty	Planner	
1	MS005	4/12/95	101.266	MS	
1	MS005	11/12/95	151.898	MS	
1	MS005	18/12/95	202.532	MS	
1	MS005	22/12/95	50.633	MS	
1	MS005	1/01/96	90.164	MS	
1	MS005	5/02/96	90.164	MS	
1	MS005	4/03/96	108.197	MS	
1	MS005	1/04/96	108.196	MS	
1	MS005	6/05/96	135.246	MS	
1=Confirm 3=Display All Outputs					
F3=Exit F8=Update F12=Previous					
09-04 SF MK KS IM II KENB20A PET166S2					

Fields:

Sel

Select. You can enter the following options:

- 1 - Confirm.
- 3 - Display All Outputs.

Schedule Quantity

The scheduled quantity can be manually amended.

Create Work Station Schedule



16/MRP

This procedure controls the transfer of suggested operation schedules from the planning system to the schedule execution system.

MRP has developed suggested operation schedules for each item planned as a scheduled item, including any MPS items within the MRP model. Each of these suggested operations carries a due time/date which is the time/date by which the scheduled quantity is to be completed through the operation.

It is the function of the Create Work Station Schedule procedure to plan how the suggested operation quantity is to be scheduled at the supporting Work Stations.

The logic used to plan suggested daily work station schedules involves back scheduling from the MRP developed operation due time/date and single threading to the shift profile of the supporting work station.

Back-schedule logic plans an operation backward from the MRP operation due time and date to determine the start time and date of the operation. The shift pattern is based on the shift profile, if defined, or the work stations shifts and standard lengths. The single threading logic refers to the shift pattern of the supporting work station for the date into which the operation is being scheduled as the maximum hours to schedule on that date.

In this manner, an operation might be scheduled over multiple days. This determination is made independently of other operations which have been or are scheduled at the work station. This means that the resulting schedules may reflect overloads planned at a work centre.

Unlike planning applications, work station schedules are never scheduled to dates earlier than the current date.



Select Create Work Station Schedule on the menu to display the first Create Work Station Schedule screen.

Create Work Station Schedule Screen 1



Select Create Work Station Schedule on the menu to display the first Create Work Station Schedule screen.

Fields:

Model

Enter the relevant planning model.



Press **Enter** on the first Create Work Station Schedule screen to display the second screen.

Create Work Station Schedule Screen 2



Press **Enter** on the first Create Work Station Schedule screen to display the second screen.

MR230	Z1 - GB Demonstration Company	User GJBAMS	8/12/95 14:08:37
Create Work Station Schedule			
Organisational Model....: 01 Organisational Model 01			
Start Date..... <u>8/12/95</u>			
To Date..... <u>8/12/95</u>			
Schedule Quantities..... _ (Blank=Integer, 1=Decimal)			
F3=Exit F8=Confirm F12=Previous			
09-42 SF MK KS IM II KENB20A PET166S2			

Fields:

Start Date

This date establishes the earliest date for which suggested daily work station schedules are created or refreshed. If you have run MRP in a net change or selective mode, only items which have been replanned replace existing suggested daily work station schedules.

New suggested daily work station schedules which fall inside the current Firm Period time fence DO NOT change firm schedules. The default for this parameter is the current date.

To Date

This date establishes the last date for which suggested daily work station schedules are created or refreshed. This date should be at least as far into the future as you intend to firm work station schedules. Set this date no later than the horizon needed for tactical capacity planning.

Schedule Quantities

Leave blank for integer, Enter 1 for decimal quantities.



Note: When MRP is run in net change or selective mode, only the operation schedules for items included in the planning are refreshed. However, if you subsequently run Create Work Station Schedules using a To Date which is later than that used when daily work station schedules were last created, additional daily work station suggested schedule operations may be created for items not included in the current MRP run.

*These relate to suggested schedule operations
within the extended To Date interval.*

MRP Report



21/MRP

This is the same as MPS Report for MPS (see MPS Report).



Select MRP Report on the menu.

For more information see MPS Report in Master Production Scheduling.



Note: MRP report options include the selection of cell ranges for cell based MRP runs.

MRP Planner Action Report



22/MRP

This is the same as MPS Planner Action Report for MPS.



Select MRP Planner Action Report on the menu.

For more information see MPS Planner Action Report in Master Production Scheduling.



Note: MRP report options include the selection of cell ranges for cell based MRP runs.

MRP Valuation Report



23/MRP

This is the same as MPS Valuation Report for MPS.



Select MRP Valuation Report on the menu.

For more information see MPS Valuation Report in Master Production Scheduling.



Note: MRP report options include the selection of cell ranges for cell based MRP runs.

MRP Reporting Profile



24/MRP

This is the same as MPS Reporting Profile for MPS.



Select MRP Reporting Profile on the menu.

For more information see MPS Reporting Profile in Master Production Scheduling.



Note: MRP report options include the selection of cell ranges for cell based MRP runs.